

PRAJWAL CHAUHAN

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Education

New York University, Tandon School of Engineering

Sep 2023 - Exp. June 2027

Doctor of Philosophy (Ph.D.) in Transportation Systems

- Developed neural operator for scientific Machine Learning (ML), spatiotemporal ML and optimization.
- Ongoing collaboration in projects related to Reinforcement Learning and Representation Learning.
- **Relevant Coursework:** Machine learning, Computational statistics, Reinforcement learning, Probability and stochastic processes, Optimization methods, Transportation economics and finance.

Indian Institute of Technology, Kanpur

Aug 2020 - Sept 2022

Master of Technology (M.Tech.) in Transportation and Highway Engineering

GBPIET, Pauri

Aug 2016 - Oct 2020

Bachelor of Technology (B.Tech.) in Civil Engineering

Work Experience

Research Engineer Intern, World Model Team, [Analog](#)

Jun 2026 – Aug 2026

- Architected a production Java/Spring Boot system that converted natural-language questions to Kusto Query Language (KQL) on Azure Data Explorer, replacing manual KQL authoring for large-scale traffic analysis.
- Reduced traffic-query latency across billions of ADX records from frequent timeouts to under 10 seconds by building a pre-aggregated graph representation that avoided raw-data scans.
- Automated data-quality checks and hypothesis testing with PySpark on Databricks, detecting data anomalies; supported intervention simulations and built a dashboard for rapid review.
- Enabled real-time, citywide congestion visualization in extended reality (XR) by designing a road-constrained interpolation algorithm aligned with city's road network.

Junior Research Fellow, Indian Institute of Technology, Tirupati

Nov 2022 – May 2023

- Conducted Traffic-state estimation (Physics informed ML) on UTD-19, US and German traffic dataset.
- Achieved accuracy equivalent to Adaptive smoothing method with significantly lower compute cost.

Technical Skills

- **Languages:** Python, Java, Kusto Query Language (KQL), MATLAB, R
- **Machine Learning and Scientific Computing:** PyTorch, TensorFlow, Jupyter, Matplotlib
- **Data and Cloud:** PySpark, Databricks, Azure Data Explorer (ADX), Graph Analytics
- **Engineering:** Spring Boot, React, Git, Docker, Kubernetes, Helm, GitHub Actions
- **Geospatial and Simulation:** OSRM, OpenStreetMap, Leaflet, Simulation of Urban MObility (SUMO)

Publications and selected Talks

- [Journal] Chauhan, P., Choutri, S. E., & Jabari, S. E. (2026). *Efficient Dilated Squeeze and Excitation Neural Operator for Differential Equations*. *Transactions on Machine Learning Research (TMLR)*. [TMLR2026](#). *Contribution*— Proposed Dilated Squeeze-and-Excitation Neural Operator, demonstrating up to $\sim 20\times$ faster inference than prior neural-operators and transformer baselines.
- [Journal] Chauhan, P., Choutri, S. E., Ghattassi, M., Masmoudi, N., & Jabari, S. E. (2025). *Neural operators struggle to learn complex PDEs in pedestrian mobility: Hughes model case study*. *Artificial Intelligence for Transportation*, 1, 100005. [AI4Transportation2025](#). *Contribution*— Built Hughes PDE benchmarks using Godunov and Wavefront Tracking Algorithm, implemented multiple neural operator baselines, designed experiments/ablations, and led evaluation.
- [Journal] Chauhan, P., Kanagaraj, V., & Asaithambi, G. *Understanding the mechanism of lane changing process and dynamics using microscopic traffic data*. *Physica A* (2022). [PhysicaA2022](#). *Contribution*— Curated microscopic trajectory dataset NGSIM, engineered lane-change features, classified lane changes, and ran statistical validation.
- [Workshop and Conference] Choutri, S. E., Chauhan, P., & Jabari, S. E. (2025). *Monte Carlo Type Neural Operator for One-Dimensional Differential Equations*. *ML4PS Workshop @ NeurIPS 2025; ICMLA 2025* [ML4PS@Neurips](#). *Contribution*— Co-developed MCNO; authored the original codebase and conducted

the experiments.

- **[Conference]** Choutri, S. E., Djehiche, B., **Chauhan, P.**, & Jabari, S. E. (2025). *Backpressure-based Mean-field Type Game for Scheduling in Multi-Hop Wireless Sensor Networks*. *EUSIPCO 2025*. [IEEE Xplore](#).
Contribution— Developed the full simulation, co-developed the theory, and executed all experiments.
- **[Talks]** *Neural operators struggle to learn complex PDEs in pedestrian mobility: Hughes model case study*, presented at: 8th Transportation Symposium (NYUAD, NY campus), UAE GSRC 2025, Clinical AI Lab (NYU Abu Dhabi), INFORMS 2025 Annual Meeting.

Positions & Achievements

- Awarded the Global Ph.D. Student Fellowship to conduct coursework and research in NYC and Abu Dhabi.
- Reviewer for AAAI 2026 Conference, AI for transportation (Journal).
- Session chair for “Sustainable and Resilient Transportation Systems” at INFORMS Annual meeting 2025.
- GATE 2020: AIR 1641/125974; TOEFL: 105/120, GRE: 319/340